

Application of a soil moisture diagnostic equation for estimating root-zone soil moisture in arid and semi-arid regions



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SUMMARY

Knowledge of soil moisture in the root zone is critical for crop growth estimation and irrigation scheduling. In this study, a soil moisture diagnostic equation is applied to estimate soil moisture at depths of 0–100 cm (because the majority of crop roots are in the top 100 cm of soil) at four USDA Soil Climate Analysis Network (SCAN) sites in arid and semi-arid regions: TX2105 in northwest Texas, NM2015 and NM2108 in east New Mexico, and AZ2026 in southeast Arizona. At each site, a dataset of 5–6 years of records of daily soil moisture, daily mean air temperature, precipitation and downward solar radiation is compiled and processed. Both the sinusoidal wave function of day of year (DOY) and a linear function of the potential evapotranspiration (PET) are used to approximate the soil moisture loss coefficient. The first four years of data are used to derive the soil moisture loss function and the empirical parameters in the soil moisture diagnostic equation. The derived loss function and empirical parameters are then applied to estimate soil moisture in the last fifth or sixth year at each site. Root mean square errors (RMSEs) of the estimated volumetric soil moistures in five different soil columns (i.e., 5 cm, 10 cm, 20 or 30 cm, 50 cm, and 100 cm) are less than 3.2 (%V/V), and the accuracy of the estimated soil moistures using the sinusoidal soil moisture loss function is slightly better than the PET-based loss functions. In addition to the three advantages of this soil moisture diagnostic equation, i.e., (1) non-cumulative errors in the estimated soil moisture, (2) no regular recalibration is required to correct the cumulative errors, and (3) no numerical iteration and initial moisture inputs are needed since only precipitation data are required, this study also demonstrates that the soil moisture diagnostic equation not only can be used to estimate surface soil moisture, but also the entire root-zone soil moisture.

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1. Introduction

Irrigation is critical for semi-arid and arid agriculture, enabling doubling or quadrupling of crop yields compared to non-irrigated production (Howell, 2001). In the semi-arid and arid regions where groundwater is the primary irrigation source, to meet growing production demands resulting from the expanding food market, both the irrigated agricultural acreage and groundwater demands for irrigation are expected to increase (Colaizzi et al., 2009), with projected increases in groundwater withdrawal as a result. Simultaneously, extremely limited precipitation in arid and semi-arid regions results in a slow recharge to the local groundwater systems. Such conditions are expected to induce a significant decline in groundwater levels and thus threaten the sustainability

of agricultural industry in these regions. Therefore, efficient water management technology is critical for sustainable agriculture (Nieswiadomy, 1985; Kim et al., 1989). A relatively accurate irrigation scheduling is vitally important to improve water use efficiency (WUE) (Stanhill, 1986). For example, estimates show that irrigation scheduling could save about 15–35% of the water normally consumed through pumping groundwater for center-pivot (20% for gated-pipe surface) irrigation systems in the Great Plains Region (e.g., Gilley and Supalla, 1983).

Most irrigation scheduling methods can be classified as either soil moisture-based or plant-based (Jones, 2004). Although plant growth directly depends on plant water status and thus a plant-based method could be more accurate than a soil moisture-based method (Jones, 2004), difficulties in measuring plant water stress in an automated fashion make the plant-based irrigation scheduling expensive to implement (Jones, 2004). On the other hand, the plant-based methods mainly time irrigations and do not provide

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information on the amounts of water to apply. Therefore, the soil moisture-based irrigation scheduling methods will continue to be popular and commonly used.

Two types of approaches currently determine the moisture content in soils for the soil moisture-based irrigation scheduling (Jones, 2004): (1) direct soil moisture measurements (e.g., Campbell and Campbell, 1982; Topp and Davis, 1985; Coopersmith et al., 2014; Susha Lekshmi et al., 2014) and (2) soil water balance calculations (e.g., Allen et al., 1998). Direct soil moisture measurement is easy to apply and relatively accurate, but is expensive to implement since soil heterogeneity requires multiple soil moisture sensors to capture the spatial variation of soil moisture (e.g., Pan and Peters-Lidard, 2008). Soil water balance calculations are also easy to apply, but are not as accurate as direct measurement and require a regular recalibration since errors in the estimated soil moisture are cumulative (Jones, 2004), although most models set a soil moisture dynamic range between the field capacity and the permanent wilting point, or between porosity and residual soil moisture content. One possible reason for the cumulative errors in the estimated soil moisture is that the soil moisture dynamics in the vadose zone (i.e., unsaturated soil) is governed by highly non-linear processes and interactions and thus are very complicated (e.g., Bastiaanssen et al., 2007). On the other hand, the uncertainties associated with soil hydraulic properties (e.g., Heathman et al., 2003) and macropores in soils (e.g., Ahuja et al., 1993, 1995) make modeling and prediction of soil moisture even more challenging. Yet, as summarized by Bastiaanssen et al. (2007), over the last 25 years a large number of unsaturated zone models have been developed (e.g., Toksoz and Kirkham, 1971; Neuman et al., 1974; Zaradny and Feddes, 1979; McLin and Gelhar, 1979; Feddes et al., 1988; Workman and Skaggs, 1989; Clemente et al., 1994; Lorre et al., 1994; Faria et al., 1994; Bastiaanssen et al., 1996; Ahuja et al., 1999; Ma et al., 2000; Ines and Droogers, 2001; Gao et al., 2013; Kumar et al., 2013a, 2013b, 2014; and others). Significant progress has been made in the modeling of the relation of soil moisture to irrigation and drainage. However, despite such promising progress, the application of these models for irrigation scheduling is almost nonexistent (Bastiaanssen et al., 2007).

Four possible barriers may be hampering the implementation of soil moisture modeling in operationalizing soil moisture-based irrigation scheduling: (1) soil moisture modeling is complex and time consuming; (2) soil moisture modeling requires regular recalibration to avoid cumulative errors resulting from the differential between simulated and actual soil moisture measurements (e.g., Jones, 2004); (3) uncertainties in soil hydraulic properties and macropores in soils are not accounted for; and (4) to our knowledge, while there are some crop growth and water usage models, e.g., Hybrid-Maize (<http://www.hybridmaize.unl.edu>) and SoySim (<http://soysim.unl.edu>), these models are mainly used for research and are difficult to utilize for farmers, because to run these tools, farmers need first to collect weather data. Therefore these tools have no real-time or online functionality. This paper aims to develop a simple and robust approach for estimating root zone soil moisture using a soil moisture diagnostic equation (Pan et al., 2003; Pan, 2012), and the estimated soil moisture can be used for scheduling irrigation in the future. Our research on the soil moisture-based irrigation scheduling has two phases. The first phase focuses on developing, applying, and validating the soil moisture diagnostic equation approach to estimating the root-zone soil moisture on non-irrigated and natural steppe/desert land in arid and semi-arid regions, which is presented in this paper. In the second phase, a follow-up study will focus on estimating the root-zone soil moisture in irrigated cropland in arid and semi-arid regions.

Pan et al. (2003) derived and validated a daily surface soil moisture diagnostic equation based on a linear stochastic differential

equation suggested by Entekhabi and Rodriguez-Iturbe (1994). The estimated soil moisture is a function of the time-weighted summation of the ratio of the historical rainfall rate to the soil moisture loss coefficient (Pan et al., 2003). Using observations from three field campaigns in grassland and agricultural regions, i.e., Monsoon'90 (Schmugge et al., 1994), Washita'92 (Jackson and Le Vine, 1996), and SGP'97 (Jackson et al., 1999), Pan et al. (2003) illustrated that the soil moisture diagnostic equation could estimate the top 5 cm soil moisture with an accuracy comparable to that of remotely sensed soil moisture. Pan (2012) applied the soil moisture diagnostic equation to four Soil Climate Analysis Network (SCAN) sites managed by Natural Resources Conservation Service (NRCS) of the United States Department of Agriculture (USDA) (Schaefer et al., 2007) and used a sinusoidal wave function of the Day of Year (DOY) to represent the soil moisture loss coefficient. The observed soil moisture data collected at these four SCAN sites were applied to test the proposed method. Small errors (root mean square error < 5%V/V) and high sample correlation coefficients ($r > 0.89$) between the observed and estimated soil moisture indicated three advantages associated with the approach, i.e., (1) non-cumulative errors in the estimated soil moisture, (2) no regular recalibration is required to correct the cumulative errors, and (3) no numerical iteration and initial moisture inputs are needed (Pan, 2012). Thus, the daily surface soil moisture diagnostic equation approach is more efficient than any other known soil moisture numerical modeling approach.

Although the soil moisture diagnostic equation has three advantages as mentioned above, this equation has only been applied to estimate surface soil moisture (Pan et al., 2003; Pan, 2012). To utilize the soil moisture diagnostic equation to estimate soil moisture for scheduling irrigation, we must estimate soil moisture in the entire root zone rather than the top 5 cm to 10 cm soils, because soil moisture in the root zone is critical for seed germination and crop growth especially during the early stage of the plant growth, and thus root-zone soil moisture is one of key variables for crop growth estimation, irrigation scheduling, and crop yield prediction and modeling. Therefore, the goal of this paper is to demonstrate the ability of the soil moisture diagnostic equation to estimate root zone soil moisture and describe the methodology of applying the soil moisture diagnostic equation to estimate root zone soil moisture. The arrangement of this paper is as follows. Section 2 describes the methodology for the derivation of the daily diagnostic soil moisture equation and the determination of the soil moisture loss function and empirical parameters in the soil moisture diagnostic equation. Section 3 introduces study sites and data. Section 4 presents results and discussion. Section 5 summarizes the findings.

2. Methodology

2.1. Derivation of a soil moisture diagnostic equation

Pan et al. (2003) and Pan (2012) derived a daily soil moisture diagnostic equation based on a linear stochastic differential equation suggested by Entekhabi and Rodriguez-Iturbe (1994). For readers who are not familiar with this equation, here we use the same approach developed in Pan et al. (2003) and Pan (2012) to derive a similar daily diagnostic equation for the root-zone soil moisture based on a simplified soil moisture dynamic equation given in Eq. (1):

$$z \frac{d\theta}{dt} = -\eta\theta + \gamma P \quad (1)$$

where z is the thickness of a soil column (from land surface down to depth z), θ is soil moisture of the soil column, $-\eta\theta$ is the loss of soil moisture, η is the loss coefficient, P is precipitation rate, and γ is the

infiltration coefficient representing the ratio of infiltration to rainfall. Eq. (1) states that the soil moisture time-change rate is equal to infiltration minus soil moisture loss. Vertical drainage and evaporation or evapotranspiration (ET) are two principal processes controlling soil water loss. In arid and semi-arid regions, since the water table usually is far below the root zone and the upward moisture transport from the water table to the root zone due to capillary rise and diffusion can be neglected, and thus the assumption involved in Eq. (1) is still valid, i.e., vertical drainage and evaporation or ET dominate soil water loss in arid and semi-arid regions. Therefore, Eq. (1) can be used to estimate soil moisture in the soil column from surface to the bottom of the root zone (e.g., 100–200 cm). Rearranging terms in Eq. (1) yields:

$$\frac{zd\theta}{-\eta\theta + \gamma P} = dt \quad (2)$$

For a time series of soil moisture at a point illustrated in Fig. 1, Eq. (2) can be integrated between time t_2 and t_1 as follows:

$$\int_{t_2}^{t_1} \frac{zd\theta}{-\eta\theta + \gamma P} = \int_{t_2}^{t_1} dt \quad (3)$$

where $t_2 < t_1$. For a time step less or equal to one day the loss coefficient and the infiltration coefficient can be assumed to be constants between time t_2 and t_1 . P in Eq. (3) is the observed precipitation between time t_1 and t_2 and independent of soil moisture. With the above assumptions, Eq. (3) becomes:

$$-\frac{z}{\eta_1} \ln \left[\frac{\theta_1 - \gamma P_1 / \eta_1}{\theta_2 - \gamma P_1 / \eta_1} \right] = t_1 - t_2 \quad (4)$$

where η_1 and P_1 are the loss coefficient and total precipitation between time t_1 and t_2 , respectively. Rearranging Eq. (4) yields:

$$\theta_1 = \theta_2 e^{-\frac{\eta_1}{z}(t_1 - t_2)} + \frac{\gamma P_1}{\eta_1} \left[1 - e^{-\frac{\eta_1}{z}(t_1 - t_2)} \right] \quad (5)$$

For a daily time step (i.e., $t_1 - t_2 = 1$ day), Eq. (5) can be reduced to:

$$\theta_1 = \theta_2 e^{-\frac{\eta_1}{z}} + \frac{\gamma P_1}{\eta_1} \left(1 - e^{-\frac{\eta_1}{z}} \right) \quad (6-1)$$

where η_1 , P_1 , θ_1 , θ_2 are daily soil moisture loss coefficient, precipitation of day 1, soil moisture of day 1, and soil moisture of day 2 which is one day before day 1, respectively. Similarly, we have soil moisture of day 2 (θ_2) as a function of η_2 , P_2 , and θ_3 :

$$\theta_2 = \theta_3 e^{-\frac{\eta_2}{z}} + \frac{\gamma P_2}{\eta_2} \left(1 - e^{-\frac{\eta_2}{z}} \right) \quad (6-2)$$

and soil moisture of day $n-1$ (θ_{n-1}) as a function of η_{n-1} , P_{n-1} , and θ_n :

$$\theta_{n-1} = \theta_n e^{-\frac{\eta_{n-1}}{z}} + \frac{\gamma P_{n-1}}{\eta_{n-1}} \left(1 - e^{-\frac{\eta_{n-1}}{z}} \right) \quad (6-n-1)$$

Substituting Eqs. (6-2), ..., (6-n-1) into Eq. (6-1) yields:

$$\theta_1 = \theta_n e^{-\sum_{i=1}^{n-1} (\eta_i/z)} + \sum_{i=2}^{n-1} \left[\frac{\gamma P_i}{\eta_i} \left(1 - e^{-\frac{\eta_i}{z}} \right) e^{-\sum_{j=1}^{i-1} (\eta_j/z)} \right] + \frac{\gamma P_1}{\eta_1} \left(1 - e^{-\frac{\eta_1}{z}} \right) \quad (7)$$

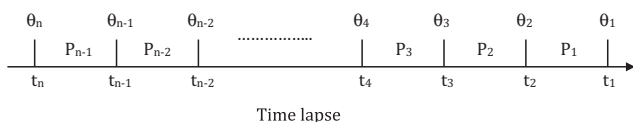


Fig. 1. Time series of soil moisture divided into $n-1$ periods; cumulative rainfall during each period is P_i , which is bounded by time t_i and t_{i+1} , θ_i is soil moisture at time t_i .

Eq. (7) indicates that as the window size (i.e., n) increases, the exponential term $e^{-\sum_{i=1}^{n-1} (\eta_i/z)}$ in the first term on the right-hand side of Eq. (7) approaches zero, and thus the contribution of the this term to θ_1 diminishes. Therefore, at a threshold time window size of n days, we can estimate soil moisture directly from a weighted average of precipitation without any information about the initial soil moisture condition as follows:

$$\theta_1 = \sum_{i=2}^{n-1} \left[\frac{\gamma P_i}{\eta_i} \left(1 - e^{-\frac{\eta_i}{z}} \right) e^{-\sum_{j=1}^{i-1} (\eta_j/z)} \right] + \frac{\gamma P_1}{\eta_1} \left(1 - e^{-\frac{\eta_1}{z}} \right) = \gamma B \quad (8)$$

where B in Eq. (8) is defined as:

$$B = \sum_{i=2}^{n-1} \left[\frac{P_i}{\eta_i} \left(1 - e^{-\frac{\eta_i}{z}} \right) e^{-\sum_{j=1}^{i-1} (\eta_j/z)} \right] + \frac{P_1}{\eta_1} \left(1 - e^{-\frac{\eta_1}{z}} \right) \quad (9)$$

and represents the summation of the weighted ratio of the rainfall rate to the loss coefficient. According to Eq. (8), as the number of days prior to day 1 increases, the contribution of precipitation on that day to the soil moisture of day 1 decreases because of the exponential decay term in Eq. (8), i.e., $e^{-\sum_{j=1}^{i-1} (\eta_j/z)}$, which ensures that the B value approaches a stable value as n increases (Pan et al., 2003; Pan, 2012).

For arid and semi-arid regions, since the annual potential evaporation or evapotranspiration (ET) rate usually is much greater than 1000 mm/yr, Pan (2012) showed that a 3-month window is sufficient for calculating soil moisture in arid and semi-arid regions using Eq. (8), without any information about the initial condition of soil moisture. In this study, a 4-month window was used for eliminating singularity in the inversion of soil moisture loss function parameters, because some arid and semi-arid sites experienced continuous non-precipitation for more than three months but less than four months.

To apply Eq. (8) to estimate soil moisture over the dynamic range of soil moisture (i.e., between residual soil moisture and saturated soil moisture), we need to determine the infiltration coefficient γ . However, in reality, the infiltration coefficient γ cannot be considered to be a single constant as it may vary with soil moisture. As B increases, soil moisture increases and approaches saturated soil moisture, and the infiltration coefficient γ will decrease and finally approach zero (Pan et al., 2003). Since, the infiltration coefficient declines as B increases, an exponential curve is the best fit of soil moisture to the B value (Pan et al., 2003). Therefore, a general form of soil moisture as a function of the B value should be:

$$\theta = \theta_{re} + (\phi_e - \theta_{re})(1 - e^{-c_4 B}) \quad (10)$$

where θ_{re} and ϕ_e are effective residual soil moisture and effective porosity, respectively, c_4 is an empirical constant related to soil hydraulic properties. Eq. (10) is called the daily soil moisture diagnostic equation. If we set z in Eq. (1) to be the thickness of the root zone, and let the loss coefficient represent the effective soil moisture loss from the entire root zone, Eq. (10) becomes the root-zone soil moisture diagnostic equation.

2.2. Soil moisture loss function

To apply the soil moisture diagnostic equation to estimate soil moisture, we must first determine the soil moisture loss coefficient that represents the ratio between soil moisture loss due to drainage and evaporation (over bare ground) or evapotranspiration (ET) (over vegetated land surface) and soil moisture. Since drainage is controlled by soil hydraulic properties and ET is related to the potential evapotranspiration (PET), the loss coefficient depends on both soil hydraulic properties (controlling drainage) and the PET (controlling the actual ET rate), while soil hydraulic properties

can be assumed as time-invariant variables and the PET varies with time because atmospheric conditions vary with time, and the loss coefficient can be approximated by a sum of one constant which is related to soil hydraulic properties and another constant multiplying PET as follows:

$$\eta = a + b \times \text{PET} \quad (11)$$

where a (in the same units as PET) and b (dimensionless) are two positive constants hereafter referred as the soil moisture loss function parameters. Eq. (11) is called the PET-based soil moisture loss function.

There are many published methods for estimating PET. Generally, these methods can be classified into three categories on the basis of data requirements (Jensen et al., 1990): (1) temperature-based methods (e.g., Thornthwaite, 1948; Hamon, 1963); (2) radiation-based methods (e.g., Jensen and Haise, 1963; McGuinness and Bordne, 1972; Priestley and Taylor, 1972); and (3) combination methods (e.g., Monteith, 1965). Oudin et al. (2005) made a comparison among 27 PET formulas and found that the method proposed by McGuinness and Bordne (1972) (hereafter the MB method) was the best in terms of streamflow simulation efficiency. In this study, we apply three PET methods in the soil moisture loss function, i.e., the Hamon method (Hamon, 1963), the JH method (Jensen and Haise, 1963) and the MB method (McGuinness and Bordne, 1972), given as follows:

(1) the Hamon method:

$$\text{PET} = \frac{2.98De_a}{T_a + 273.2} \quad (12)$$

where PET is the potential evapotranspiration rate in cm/day, D is day length in hours calculated using the formulas (Dingman, 2002) presented in Appendix A, and e_a is the saturation vapor pressure at the mean daily temperature T_a (°C) in kPa and given by:

$$e_a = 0.611 \times \exp\left(\frac{17.3T_a}{T_a + 237.3}\right) \quad (13)$$

(2) the JH method:

$$\text{PET} = \frac{R_s T_a}{0.4\rho\lambda_v} \quad (14)$$

where PET is the potential evapotranspiration rate in cm/day, R_s is downward solar radiation in MJ/(m² × day), T_a is the mean daily temperature (°C), $\rho = 1000 \text{ kg/m}^3$ is water density, and $\lambda_v = 2.5 \text{ MJ/kg}$ is the latent heat of vaporization.

(3) the MB method:

$$\text{PET} = \frac{R_s(T_a + 5)}{0.68\rho\lambda_v} \quad (15)$$

where PET is the potential evapotranspiration rate in cm/day, R_s is downward solar radiation in MJ/(m² × day), T_a is the mean daily temperature (°C), $\rho = 1000 \text{ kg/m}^3$ is water density, and $\lambda_v = 2.5 \text{ MJ/kg}$ is the latent heat of vaporization.

The potential evapotranspiration (PET) is also known as the atmospheric demand evapotranspiration and is controlled by the atmospheric conditions, e.g., solar radiation drives evaporation from bare soils and transpiration from vegetation. The water vapor gradient between the atmosphere and the land surface is controlled by air temperature and relative humidity. Wind speed controls the convection of water vapor from the land surface into the atmosphere. Canopy structure also affects the vertical profiles of air temperature and wind, and in turn influences the exchange of water and energy between the land surface and the atmosphere. Since all those atmospheric conditions mentioned above (e.g., daily

mean values of solar radiation, air temperature, and relative humidity at a location) can be approximated by functions of the DOY, Pan (2012) used a sinusoidal wave function of DOY to represent the daily soil moisture loss coefficient η (which depends on soil hydraulic properties and the PET rate) as follows:

$$\eta_i = c_1 + c_2 \sin\left[\frac{2\pi(\text{DOY}_i + c_3)}{365}\right] \quad (16)$$

where η_i is the loss coefficient of day i , DOY_i is the DOY of day i , and c_1 , c_2 , and c_3 are positive soil moisture loss function parameters. Although the sinusoidal soil moisture loss function has one more parameter than the PET-based soil moisture loss function (i.e., Eq. (11)), the advantage associated with this sinusoidal function is that neither air temperature nor solar radiation data are needed. Using the sinusoidal soil moisture loss function to estimate soil moisture at four SCAN sites, Pan (2012) showed that the root mean square errors (RMSEs) in the estimated soil moisture were less than 5.0%V/V, which indicates that the sinusoidal soil moisture loss function can capture daily variation of soil moisture loss quite well. To make a comparison between the sinusoidal and PET-based soil moisture loss functions in terms of the accuracy of the estimated soil moisture, both types of functions were tested in this study.

2.3. Determining parameters in the daily diagnostic soil moisture equation approach

There are five (for the PET-based loss function) or six (for the sinusoidal function) parameters involved in the soil moisture diagnostic equation approach, i.e., a and b in the PET-based soil moisture loss function (Eq. (11)), or c_1 , c_2 , c_3 in the sinusoidal soil moisture loss function (Eq. (16)), and effective residual soil moisture θ_{re} , effective porosity ϕ_e , and c_4 in the daily diagnostic soil moisture equation (Eq. (10)). The soil moisture loss function parameters, depending on geographic location, soil, and vegetation characteristics, can be inversely determined by maximizing the correlation coefficient between observed soil moisture and B value (i.e., best fit between observed soil moisture and B value) (Pan, 2012), given as follows:

$$\text{Max} \left\{ \frac{\sum_{i=1}^m [(\theta_i - \bar{\theta})(B_i - \bar{B})]}{\sqrt{\sum_{i=1}^m (\theta_i - \bar{\theta})^2} \sqrt{\sum_{i=1}^m (B_i - \bar{B})^2}} \right\}_{(a,b) \text{ or } (c_1, c_2, c_3)} \quad (17)$$

where θ_i and B_i , $i = 1, \dots, m$ are soil moisture measurements and computed B values, respectively, $\bar{\theta}$ and $\bar{B}$ are mean values of soil moisture measurements and computed B values, respectively. There are two commonly used methods to achieve the best fit, i.e., the Gauss–Newton method (Fletcher, 1987) or a simple “global search” method. To maximize the correlation coefficient given in Eq. (17) using the Gauss–Newton method, we need to construct a Jacobian matrix. However, two or three unknown parameters, i.e., a and b in the PET-based soil moisture loss function, or c_1 , c_2 , and c_3 in the sinusoidal soil moisture loss function and high nonlinearity of the B expression (i.e., Eq. (9)) make it difficult to obtain an analytical expression of the Jacobian matrix. Compared to the Gauss–Newton method, the global search method is simpler.

There are several methods for a global search. The simplest method is to exhaustively search the parameter space of unknown parameters. Although this method is simple and efficient for problems with few parameters, the computation becomes unfeasible for problems with many parameters and large parameter spaces. To invert a nonlinear equation with high dimensionality (i.e., many parameters), a Monte Carlo search, i.e., a random walk in the parameter space, is a simple and robust method. Since the concept of the Monte Carlo method is simple and not new, it has been widely used to solve inverse problems (Mosegaard and Tarantola, 1995),

especially in seismology (e.g., Keilis-Borok and Yanovskaya, 1967; Rothman, 1986; Landa et al., 1989; Mosegaard and Vestergaard, 1991; Koren et al., 1991). Because the main objective of this paper is to demonstrate the feasibility of the suggested method for estimating root zone soil moisture, we used the simple and feasible Monte Carlo search method here.

According to Eq. (16), c_1 represents the mean value of the soil moisture loss function, and c_2 is the magnitude of the loss function, and c_3 is the phase of the sinusoidal wave. Since the loss coefficient cannot be negative (i.e., always greater than or equal to zero), c_2 must be less than or equal to c_1 . Both c_1 and c_2 are in the same unit as precipitation (i.e., length/day, because a daily time step is used in this study) and c_3 is expressed in the DOY (between 0 and 365 or 366). According to the map of mean annual pan evaporation for the contiguous United States created by Farnsworth and Thompson (1982), the maximum annual mean pan evaporation of about 355.6 cm occurs in southeast Arizona. Considering higher pan evaporation usually happens during six months (April–September) and assuming negligible pan evaporation during other six months (January–March, October–December), the possible maximum daily pan evaporation is about $2 \times 355.6 \text{ cm}/180 \text{ day} = 3.95 \text{ cm/day}$. Therefore, the searching range for c_1 can be set between 0 and 4 cm/day, and thus the searching domain of the sinusoidal soil moisture loss function parameters is given as follows:

$$\text{Searching domain} = \{0 < c_1 < 4 \text{ cm/day}, 0 < c_2 \leq c_1, 0 \leq c_3 \leq 365\} \quad (18)$$

During the searching process, if the resulting c_1 is close to 4 cm/day, we need to expand the searching domain for c_1 . Similarly, the searching domain of the PET-based soil moisture loss function parameters is given as follows:

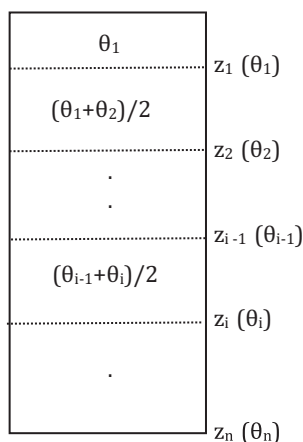


Fig. 2. Soil moisture is measured at different depths, z_i ($i = 1, \dots, n$).

$$\text{Searching domain} = \{0 < a < 4 \text{ cm/day}, 0 < b < 40\} \quad (19)$$

If the resulting a is close to 4 cm/day or b is close to 40, we need to expand the searching domain for a or b .

3. Study sites and data

The Soil Climate Analysis Network (SCAN), a comprehensive, nationwide soil moisture and climate information system, is administrated by the United States Department of Agriculture Natural Resources Conservation Service (USDA NRCS) through the National Water and Climate Center (NWCC), in cooperation with the NRCS National Soil Survey Center (NSSC) (Seyfried et al., 2005; Schaefer et al., 2007). The SCAN system measures soil moisture content hourly at depths of 5, 10, 20, 30, 50, and 100 cm. Some SCAN sites also measure soil moisture at 150 and 200 cm. To apply the soil moisture diagnostic equation proposed in this study for estimating root zone soil moisture, root zone soil moisture data are needed. However, entire root zone soil moisture is not measured automatically and directly at the SCAN sites; rather, it can be calculated from soil moisture measurements at different depths, from the top level z_1 down to the bottom of the root zone z_n (see Fig. 2), using the following expression:

$$\theta = \frac{z_1 \theta_1 + \sum_{i=2}^n [(z_i - z_{i-1})(\theta_{i-1} + \theta_i)/2]}{z_n} \quad (20)$$

where θ_i is soil moisture measurement at depth of z_i , $i = 1, \dots, n$, and θ represents the mean soil moisture in the soil column from the surface (i.e., $z = 0$) to the depth of z_n .

Since the main objective of this study is to develop an approach to estimate root zone soil moisture which is extremely important for arid and semi-arid agricultural production regions, four SCAN sites (see Table 1) on steppe/desert lands in New Mexico, southeast Arizona, and the Texas High Plains region were chosen to demonstrate the ability of the soil moisture diagnostic equation to estimate root zone soil moisture and illustrate the methodology described in Section 2. At each site, soil moistures in five soil columns were calculated using Eq. (20). These five different soil column thicknesses at each site were selected from 8 possible measurement depths with few missing data and listed in Table 1. In addition to soil moisture measurements, daily precipitation and daily mean air temperature collected at these SCAN sites were also downloaded from the SCAN website (<http://www.wcc.nrcs.usda.gov/scan>). In the JH and MB PET methods, in addition to daily mean air temperature, daily solar radiation is also needed for computing PET. However, at four selected SCAN sites, solar radiation data are not available during the study period. The daily downward global solar radiation in the NCEP/NCAR reanalysis data (Kalnay et al., 1996) were used and the grid center of each grid cell covering one of four selected SCAN sites is also listed in Table 1.

Table 1
Four SCAN sites.

Site ID	Site name	State	Lat.	Lon.	Land-cover	NCEP/NCAR reanalysis grid center (lon, lat)	Soil moisture measurement depths (cm)	Parameter-estimation period	Model-testing period
TX2105	Levelland	TX	33°33'N	102°22'W	Steppe	(256.875°, 33.3328°)	5, 10, 30, 50, 100	3/12/2005–11/27/2008 10/15/2009–12/31/2010	1/1/2011–12/31/2011
NM2015	Adams Ranch	NM	34°15'N	105°25'W	Steppe	(255°, 33.3328°)	5, 10, 20, 50, 100	1/1/2005–11/30/2006	1/1/2010–9/15/2011
NM2108	Willow Wells	NM	33°32'N	103°38'W	Steppe/Desert	(256.875°, 33.3328°)	5, 10, 20, 50, 100	1/1/2007–8/3/2009 3/13/2005–12/8/2008	1/1/2011–12/31/2011
AZ2026	Walnut Gulch	AZ	31°44'N	110°3'W	Steppe	(249.375°, 31.4281°)	5, 10, 20, 50, 100	8/6/2009–12/31/2010 6/4/2004–3/28/2007 6/30/2007–12/31/2008	1/1/2009–12/31/2009

When we apply the soil moisture diagnostic equation to estimate soil moisture, the first step is calibration, i.e., using at least two years of continuous observations of daily soil moisture and daily precipitation to determine soil moisture loss function parameters and three empirical parameters in the soil moisture diagnostic equation as described in Sections 2.2 and 2.3. Since there are some missing data gaps for all four selected SCAN sites, at each site we chose two periods with continuous soil moisture and other atmospheric forcing data records as the parameter estimation period (PEP) (or calibration period), during which the observed soil moisture data are used to determine the soil moisture loss function parameters, effective hydraulic properties and the empirical coefficient in the soil moisture diagnostic equation. The total length of the PEP for each site is about four years (see Table 1). Another

one-year period of data record is used for validation (see Table 1) and this one-year period is called the model-testing period (MTP), during which the derived parameters in the parameter estimation period are applied to estimate soil moisture and compare it to the observed soil moisture in the MTP.

Time series plots of daily downward solar radiation, daily mean temperature, daily precipitation, and observed daily soil moistures in five soil columns during the parameter estimation period (PEP) and the model-testing period (MTP) at TX2105, NM2015, and AZ2026 are plotted in Figs. 3–6, respectively. The soil particle size distribution (PSD) at each site in different soil layers surveyed by USDA is listed in Table 2. Corresponding to soil moisture measurement depths, effective soil PSDs were also calculated for each soil column and shown in Table 3.

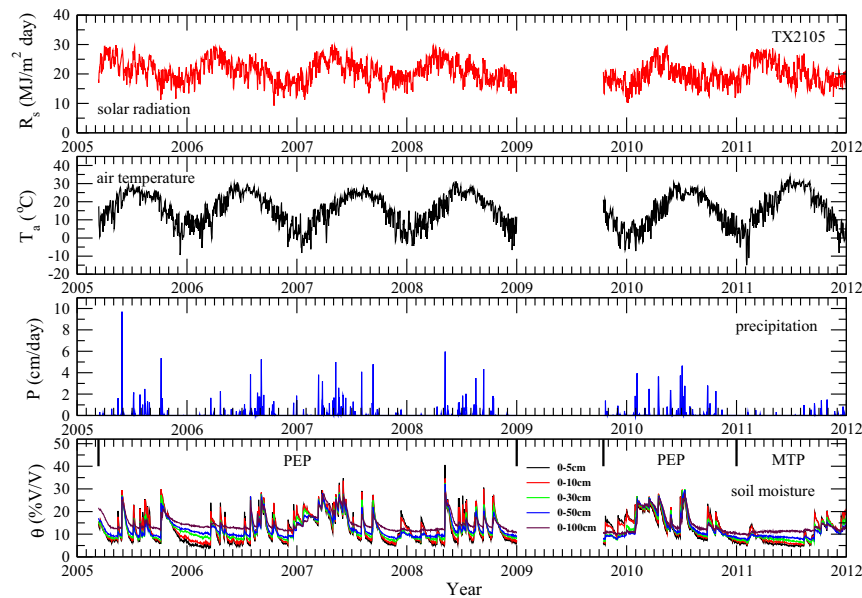


Fig. 3. Daily downward solar radiation, daily mean temperature, daily precipitation, and observed daily soil moistures in five soil columns during the parameter estimation period (PEP) and the model-testing period (MTP) at TX2105.

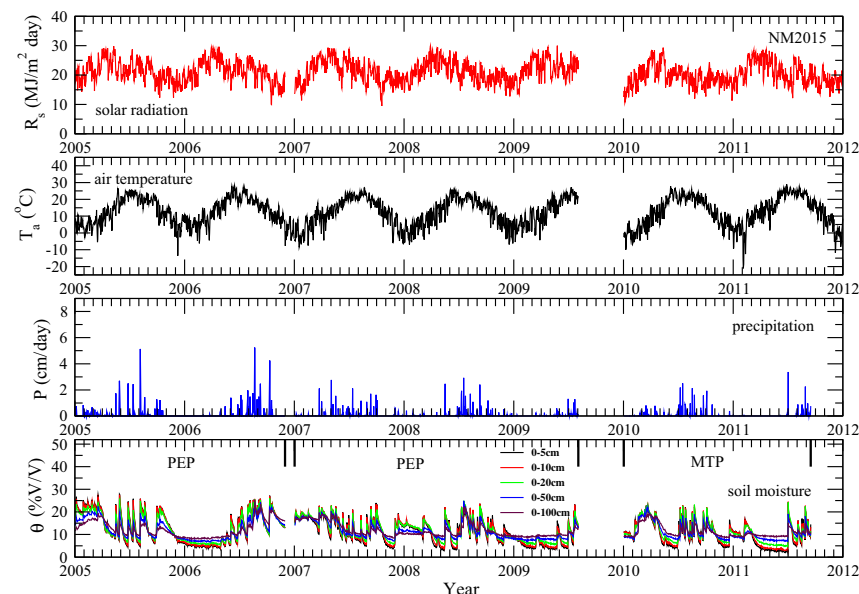


Fig. 4. Daily downward solar radiation, daily mean temperature, daily precipitation, and observed daily soil moistures in five soil columns during the parameter estimation period (PEP) and the model-testing period (MTP) at NM2015.

4. Results and discussion

4.1. Results

Using the Monte Carlo global searching method described in Section 2, we determined the optimal soil moisture loss function for each soil column at each of four SCAN sites. The results are listed in Table 4. According to Table 4, all determined soil moisture loss coefficient parameters produced high sample correlation coefficients (all $r > 0.7$) between the observed soil moistures and computed B values, which indicates that both the sinusoidal (as a function of DOY) and the PET-based soil moisture loss functions work well not only in the top surface soil layer but also in the

entire root zone. However, the sample correlation coefficient created by the sinusoidal loss function is slightly higher than the PET-based soil moisture loss function. One possible reason is that the sinusoidal loss function has one more parameter than the PET-based loss function. The scatter plots of observed soil moisture versus computed B values using the sinusoidal soil moisture loss function are shown in Fig. 7. The Matlab curve fitting toolbox (i.e., the nonlinear least squares method) was used to best fit the scatter plots using the soil moisture diagnostic equation (i.e., Eq. (10)) and the empirical coefficient c_4 , effective residual soil moisture (θ_{re}) and effective porosity (ϕ_e) in the soil moisture diagnostic equation were determined for each soil column at each of four SCAN sites (see Table 5). The best-fit curves were also plotted in

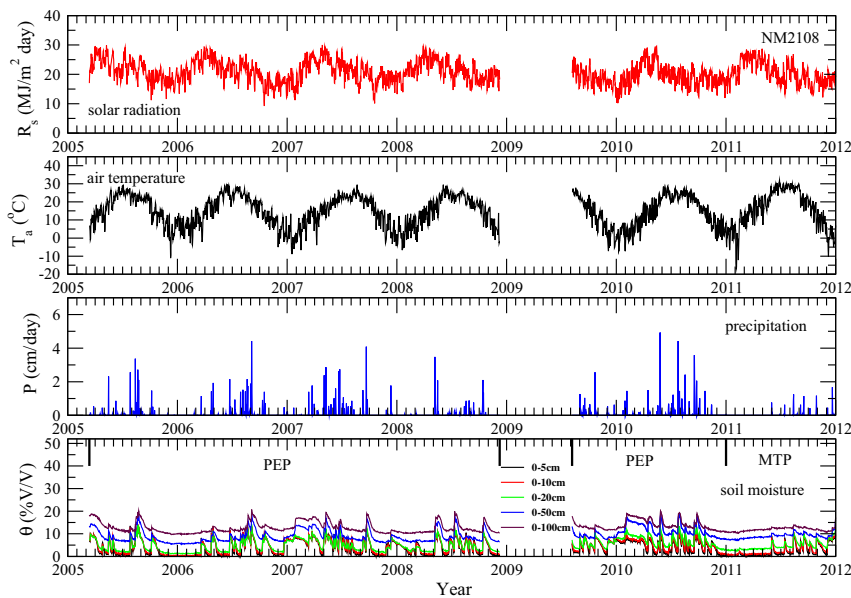


Fig. 5. Daily downward solar radiation, daily mean temperature, daily precipitation, and observed daily soil moistures in five soil columns during the parameter estimation period (PEP) and the model-testing period (MTP) at NM2108.

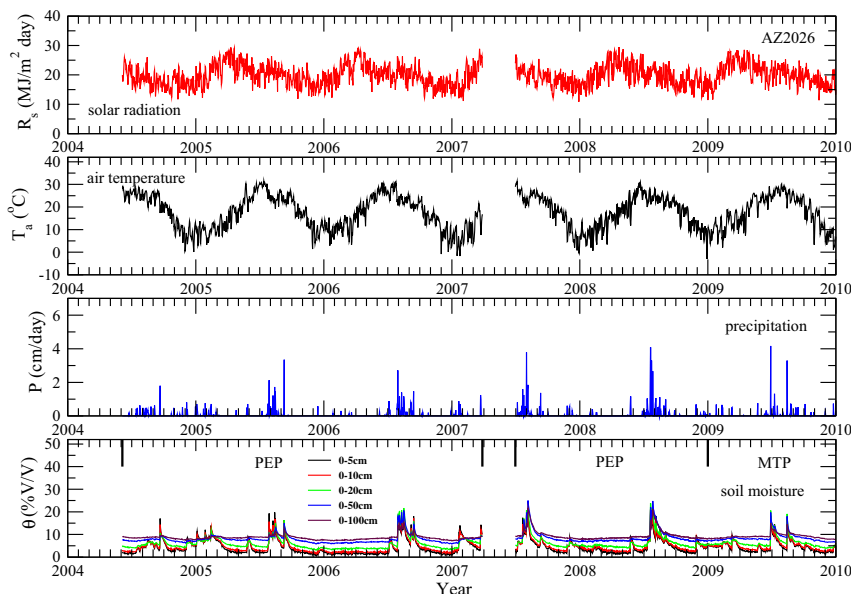


Fig. 6. Daily downward solar radiation, daily mean temperature, daily precipitation, and observed daily soil moistures in five soil columns during the parameter estimation period (PEP) and the model-testing period (MTP) at AZ2026.

Table 2

Soil Particle Size Distributions (PSD) at four SCAN sites.

TX2105			NM2015			NM2108			AZ2026		
Depth (cm)	Clay (%)	Sand (%)	Depth (cm)	Clay (%)	Sand (%)	Depth (cm)	Clay (%)	Sand (%)	Depth (cm)	Clay (%)	Sand (%)
0–15	29.8	40.4	0–5	7.5	76.7	0–24	5.5	90.4	0–7	11.7	70.0
15–28	12.7	69.2	5–18	11.6	71.6	24–46	18.1	78.0	7–32	22.4	48.2
28–41	16.8	68.3	18–61	19.1	61.1	46–87	22.1	74.1	32–52	21.0	40.4
41–60	22.6	60.9	61–81	17.3	62.3	87–152	11.3	73.8	52–81	15.6	45.1
60–88	26.1	56.4	81–104	14.5	67.8				81–95	12.4	63.2
88–140	20.5	65.5							95–110	8.0	75.0
140–203	17.5	61.0									

Table 3

Computed soil PSD in each soil column at four SCAN sites.

TX2105				NM2015				NM2108				AZ2026			
Depth (cm)	Clay (%)	Sand (%)	ϕ^* (%V/V)	Depth (cm)	Clay (%)	Sand (%)	ϕ^* (%V/V)	Depth (cm)	Clay (%)	Sand (%)	ϕ^* (%V/V)	Depth (cm)	Clay (%)	Sand (%)	ϕ^* (%V/V)
0–5	29.8	40.4	43.8	0–5	7.5	76.7	39.2	0–5	5.5	90.4	37.5	0–5	11.7	70.0	40.1
0–10	29.8	40.4	43.8	0–10	9.6	74.2	39.6	0–10	5.5	90.4	37.5	0–10	14.9	63.5	40.9
0–30	21.5	54.7	42.0	0–20	11.3	71.8	39.9	0–30	8.0	87.9	37.8	0–20	18.7	55.8	41.9
0–50	20.7	58.8	41.5	0–50	16.0	65.4	40.7	0–50	12.4	83.6	38.4	0–50	20.4	48.4	42.8
0–100	20.5	61.0	41.2	0–100	16.3	64.8	40.7	0–100	15.8	78.8	39.0	0–100	17.3	50.7	42.5

* Based on Cosby et al. (1984) pedotransfer function: $\phi = 48.9 - 0.126 \times \text{sand} (\%)$.**Table 4**

Optimal soil moisture loss function parameters.

Site	Depth	Sinusoidal loss function				Hamon PET			JH PET			MB PET		
		c_1	c_2	c_3	$r_{0,B}$	a	b	$r_{0,B}$	a	b	$r_{0,B}$	a	b	$r_{0,B}$
TX	0–5	0.375	0.315	262	0.88	0.001	1.717	0.87	0.020	1.207	0.88	0.004	1.848	0.86
	2	0.729	0.628	261	0.90	0.004	3.301	0.89	0.025	2.234	0.90	0.006	3.227	0.88
	1	1.603	1.468	255	0.92	0.003	7.812	0.90	0.020	5.570	0.92	0.002	7.975	0.89
	0	2.280	2.174	250	0.92	0.001	11.845	0.89	0.001	8.534	0.90	0.001	11.845	0.86
	5	3.827	3.741	239	0.93	0.021	17.779	0.84	0.002	13.878	0.85	0.021	17.779	0.81
NM	0–5	0.294	0.257	259	0.87	0.001	1.717	0.86	0.020	1.207	0.87	0.004	1.848	0.86
	2	0.538	0.487	257	0.88	0.001	3.544	0.87	0.033	2.539	0.88	0.006	3.227	0.87
	0	1.023	0.950	252	0.91	0.003	6.835	0.89	0.045	4.756	0.90	0.010	6.651	0.89
	1	1.917	1.721	247	0.94	0.001	13.151	0.94	0.255	8.679	0.94	0.001	12.169	0.93
	5	2.389	2.251	242	0.93	0.007	17.578	0.94	0.658	10.764	0.93	0.002	16.864	0.93
NM	0–5	0.539	0.470	250	0.79	0.006	2.794	0.76	0.040	1.926	0.78	0.010	2.782	0.76
	2	1.021	0.900	251	0.82	0.001	4.977	0.79	0.028	3.631	0.81	0.001	5.255	0.79
	1	1.743	1.570	244	0.85	0.010	7.902	0.79	0.014	6.287	0.83	0.001	8.917	0.80
	0	3.178	3.135	237	0.83	0.002	16.394	0.73	0.001	11.199	0.77	0.006	15.733	0.74
	8	3.905	3.846	235	0.80	0.003	27.323	0.70	0.048	17.932	0.71	0.002	25.690	0.70
AZ	0–5	0.429	0.237	262	0.93	0.007	1.607	0.93	0.020	1.207	0.92	0.010	1.616	0.92
	2	0.712	0.378	254	0.94	0.047	2.622	0.94	0.062	1.886	0.94	0.010	2.782	0.93
	0	1.199	0.527	226	0.93	0.407	3.391	0.93	0.474	2.513	0.92	0.281	3.962	0.92
	2	2.774	1.536	163	0.93	3.877	0.000	0.92	3.872	0.010	0.92	3.872	0.010	0.92
	6	3.995	3.501	147	0.91	6.461	0.001	0.89	6.606	0.016	0.89	6.606	0.016	0.89

Fig. 7. During the best fitting of computed B values if the Matlab determined effective porosity was greater than the computed porosity based on the Cosby et al.'s (1984) pedotransfer function (see Table 3), the effective porosities was set to be the computed porosity based on the Cosby et al.'s (1984) pedotransfer function (see Table 5).

The time series plots of the estimated (using the sinusoidal loss function) and observed soil moistures during the parameter-estimation period are shown in Figs. 8–11. Since the difference in the estimated soil moistures between using the sinusoidal and PET-based soil moisture loss functions is insignificant, to make each plot clear, the estimated soil moistures using the PET-based loss functions are not plotted in Figs. 8–11. The RMSEs and the sample correlation coefficients (r) are listed in Table 5. All RMSEs are less than 2.65 (%V/V) and all r s are greater than 0.7, which

indicates that the proposed soil moisture diagnostic equation approach is capable of estimating root zone soil moisture from precipitation data accurately, and both the sinusoidal and PET-based soil moisture loss functions can capture the soil moisture dry-down process very well.

To test the soil moisture diagnostic equation approach, we applied the derived soil moisture loss functions and the empirical parameter c_4 , effective residual soil moisture and effective porosity in the soil moisture diagnostic equation based on the observed soil moistures in the parameter estimation period (PEP) to calculate soil moisture in the model-testing period (MTP). Results are also plotted in Figs. 8–11 and RMSEs and the sample correlation coefficients between the observed and estimated soil moistures are listed in Table 6. All RMSEs of the estimated soil moistures in the model-testing period are close to the RMSEs of the estimated soil

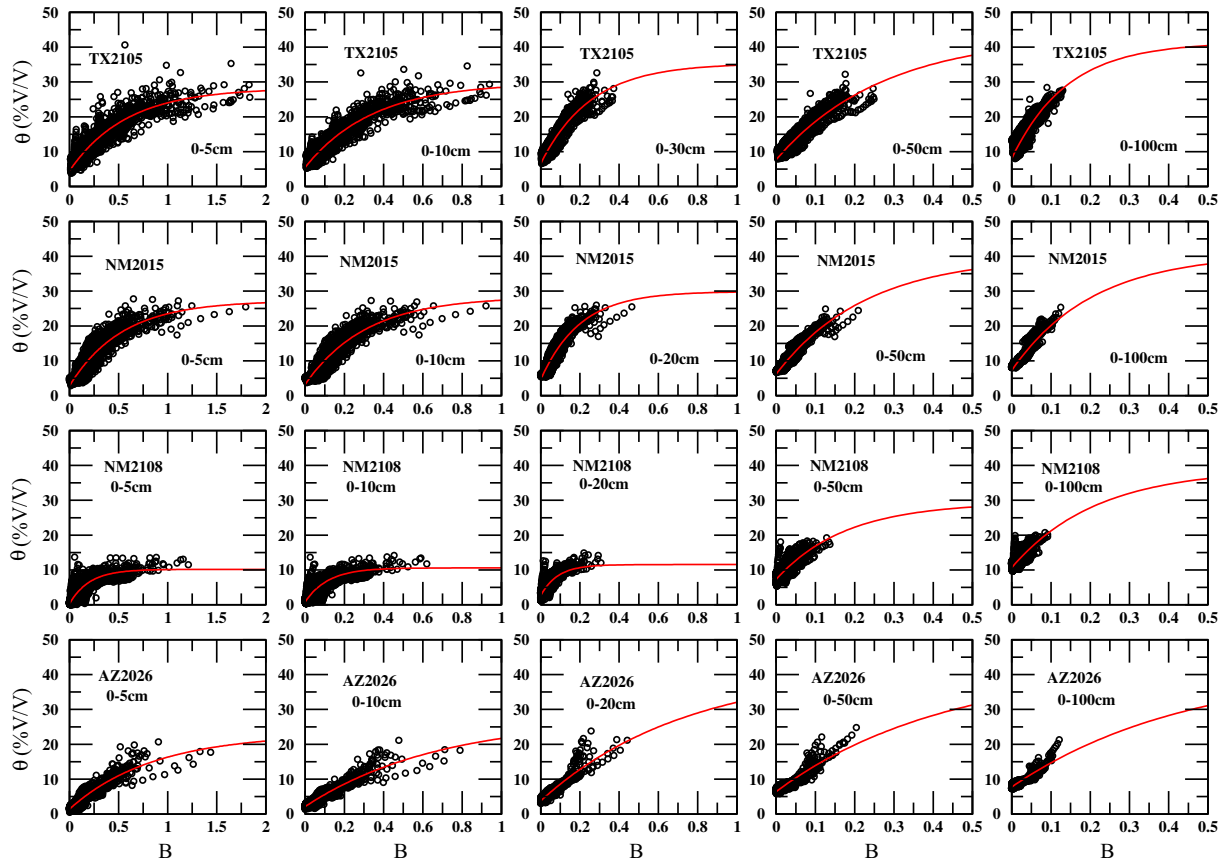


Fig. 7. Scatterplots of observed soil moisture versus computed B values, and best-fit curves (red) of the scatterplots. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

moistures in the parameter-estimation period and less than 3.2 (%V/V), which demonstrates that errors in the estimated root zone soil moisture using the soil moisture diagnostic equation are not cumulative, therefore no re-calibration of the model is needed. The sample correlation coefficients in the model-testing period are also close to those in the parameter-estimation period. Overall, our model works well, but further refinements can be considered as we discuss in Section 4.2.

4.2. Potential areas for improving root-zone soil moisture estimation

According to the results presented in Section 4.1, the accuracy of the estimated soil moistures using the sinusoidal loss function is slightly better than the PET-based loss functions. There are two possible reasons: (1) the sinusoidal loss function has one more empirical parameter than the PET-based loss functions; (2) the relation between soil moisture dry-down and PET may not be linear. Although the PET-based soil moisture loss functions only have two parameters, air temperature data is needed for the temperature-based PET methods and solar radiation data are needed for the radiation-based and combination PET methods. Therefore, this study's results do not reflect any advantages relating PET to the soil moisture loss function.

Since the focus of this paper is to demonstrate the ability of the soil moisture diagnostic equation to estimate root-zone soil moisture, here we briefly discuss the potential methods for improving accuracy of the estimated soil moisture using the sinusoidal soil moisture loss function. There are four potential improvements. First, evapotranspiration (ET) is generally positively correlated

with air temperature and solar radiation (e.g., [Thornthwaite, 1948](#); [Monteith, 1965](#); [Priestley and Taylor, 1972](#)). The sinusoidal loss function actually reflects this general relationship, i.e., during the summer (winter) seasons the loss coefficient is high (low). However, the sinusoidal loss function can be thought of as a long-term averaged daily soil moisture loss condition. To consider the impacts of deviation of air temperature and solar radiation from their long-term averages on ET, we can add two terms to the sinusoidal soil moisture loss function as follows:

$$\eta_i = c_1 + c_2 \sin \left[\frac{2\pi(\text{DOY}_i + c_3)}{365} \right] + c_5 \left(\frac{T_i}{T_{ai}} \right) + c_6 \left(\frac{R_i}{R_{ai}} \right) \quad (21)$$

where c_5 and c_6 are constants, T_{ai} and R_{ai} are long-term averaged daily mean air temperature in Kelvin and solar radiation in MJ/(m² × day) of day i . T_i and R_i are air temperature and solar radiation of day i , respectively. The same procedure described in Section 2 can be used to determine c_5 and c_6 . By comparing the errors in the estimated soil moisture between conditions with and without air temperature and solar radiation included in the soil moisture loss function, we will be able to determine if there is a tradeoff between number of variables included in the soil moisture loss function and the accuracy of the estimated soil moisture. This will also determine if the inclusion of air temperature and solar radiation in the soil moisture loss function will lead to improved accuracy of the estimated soil moisture. If Eq. (21) does not improve the accuracy of the estimated soil moisture, we can also test other schemes of including wind speed and vapor pressure deficit along with air temperature and solar radiation in the soil moisture loss function based

Table 5

Effective hydraulic properties and statistics of the estimated soil moisture during the parameter-estimating period.

Site	Depth	Sinusoidal loss function					Hamon PET					JH PET					MB PET				
		c_4	θ_{re}	ϕ_e	RMSE	r	c_4	θ_{re}	ϕ_e	RMSE	r	c_4	θ_{re}	ϕ_e	RMSE	r	c_4	θ_{re}	ϕ_e	RMSE	r
TX	0–5	1.72	4.54	28.24	2.40	0.92	1.78	5.53	28.08	2.56	0.91	1.68	5.30	28.17	2.47	0.92	1.83	5.94	28.44	2.65	0.91
	2	2.90	5.49	29.79	2.05	0.94	2.88	6.41	30.02	2.27	0.92	2.53	6.03	30.61	2.18	0.93	2.62	6.56	31.08	2.42	0.83
	1	3.93	6.51	35.35	1.77	0.94	3.51	7.41	39.09	2.15	0.91	2.73	7.17	44.67	1.97	0.92	3.15	7.58	42.22	2.28	0.90
	0	3.66	7.73	43.39	1.73	0.93	4.47	8.29	41.5	2.25	0.88	4.34	7.94	41.5	2.12	0.89	4.47	8.36	41.50	2.41	0.86
	5	5.57	10.67	41.20	1.45	0.93	6.25	10.69	41.20	2.12	0.84	6.35	10.56	41.20	2.10	0.84	6.19	10.76	41.20	2.29	0.81
NM	0–5	1.88	2.85	27.24	2.27	0.92	2.51	3.79	24.42	2.20	0.92	2.16	3.62	25.49	2.18	0.93	2.62	4.26	24.74	2.22	0.92
	2	3.22	3.21	28.39	2.11	0.92	4.87	4.44	24.88	2.02	0.93	4.23	4.27	25.93	1.99	0.93	4.30	4.58	25.45	2.12	0.92
	0	5.02	4.53	29.89	1.71	0.94	7.34	5.73	25.86	1.73	0.93	6.19	5.43	27.15	1.69	0.94	7.01	5.91	26.25	1.79	0.93
	1	4.36	6.22	40.0	1.13	0.95	6.67	6.99	31.64	1.13	0.95	5.68	6.85	34.2	1.15	0.95	5.91	7.05	32.98	1.21	0.94
	5	4.87	7.22	40.7	1.15	0.92	5.32	7.93	40.7	1.10	0.93	5.17	7.90	40.7	1.15	0.92	5.19	7.99	40.7	1.17	0.92
NM	0–5	4.93	0.36	10.18	1.55	0.88	5.99	0.48	10.03	1.68	0.87	5.11	0.35	10.06	1.64	0.88	6.09	0.54	9.94	1.68	0.87
	2	8.53	0.84	10.61	1.40	0.89	9.00	0.98	10.63	1.60	0.87	7.83	0.86	10.67	1.53	0.88	9.94	1.06	10.44	1.56	0.87
	1	11.55	2.64	12.2	1.34	0.86	10.81	2.09	11.94	1.57	0.84	9.09	2.01	12.75	1.40	0.87	12.63	2.19	11.74	1.48	0.86
	0	5.65	7.33	29.38	1.59	0.79	11.05	6.70	20.40	1.94	0.75	6.47	6.45	26.03	1.83	0.78	10.97	6.66	20.28	1.92	0.76
	8	4.66	10.64	39.0	1.40	0.76	11.88	10.74	26.32	1.77	0.70	5.77	10.54	37.4	1.73	0.71	11.89	10.72	25.7	1.78	0.70
AZ	0–5	1.173	1.435	22.99	1.09	0.89	1.04	1.47	24.48	1.12	0.89	1.02	1.49	24.41	1.14	0.88	1.26	1.49	21.34	1.13	0.89
	2	1.467	2.128	27.57	0.95	0.90	1.27	2.18	30.44	0.98	0.89	1.37	2.39	31.94	1.10	0.87	1.48	2.21	26.52	1.02	0.89
	0	1.355	3.733	41.9	0.98	0.87	1.36	3.78	41.9	1.04	0.86	1.34	3.83	41.9	1.06	0.85	1.32	3.82	41.9	1.07	0.85
	2	2.298	6.434	42.8	0.89	0.84	2.59	6.61	42.8	0.95	0.82	2.59	6.61	42.8	0.95	0.82	2.59	6.61	42.8	0.95	0.82
	6	2.231	7.727	42.5	0.64	0.82	2.98	7.85	42.5	0.70	0.78	3.03	7.86	42.5	0.70	0.78	3.03	7.86	42.5	0.70	0.78

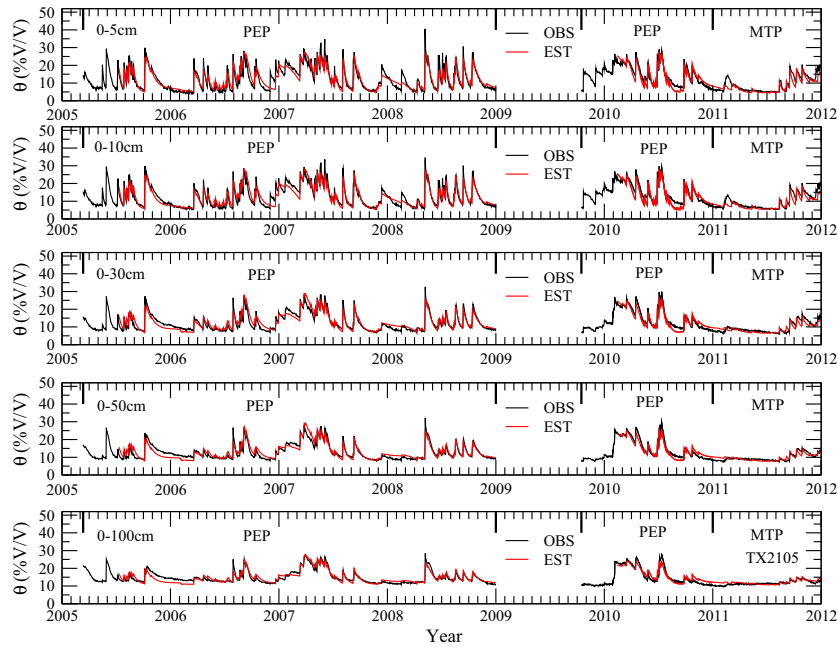


Fig. 8. Observed and estimated soil moisture at TX2105 in 0–5 cm, 0–10 cm, 0–30 cm, 0–50 cm, and 0–100 cm soil columns.

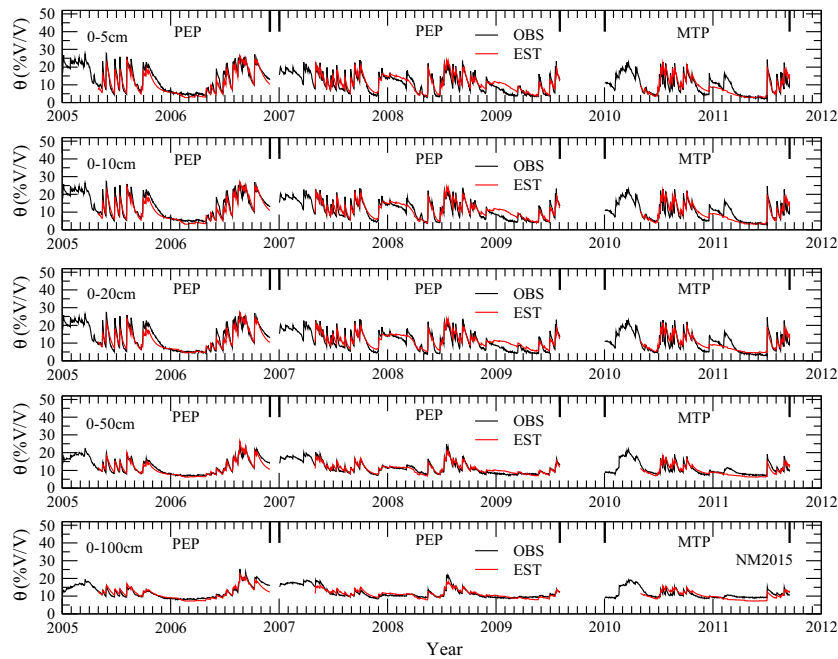


Fig. 9. Observed and estimated soil moisture at NM2015 in 0–5 cm, 0–10 cm, 0–20 cm, 0–50 cm, and 0–100 cm soil columns.

on published ET functions (e.g., [Thornthwaite, 1948](#); [Monteith, 1965](#); [Priestley and Taylor, 1972](#); [Allen et al., 1998](#)).

Secondly, although the atmospheric conditions are the dominant factors controlling ET, in vegetated areas ET also depends on vegetation height, leaf area index (LAI), stomata resistance, and other vegetation characteristics. To consider the vegetation impact on soil moisture loss, we can also include LAI in Eq. (21) as follows:

$$\eta_i = c_1 + c_2 \sin \left[\frac{2\pi(\text{DOY}_i + c_3)}{365} \right] + c_5 \left(\frac{T_i}{T_{ai}} \right) + c_6 \left(\frac{R_i}{R_{ai}} \right) + c_7 \text{LAI} \quad (22)$$

By comparing errors in the estimated soil moisture from the calculations made with and without LAI, we will be able to determine during which plant growth stages including LAI in the soil moisture loss function the improvement in the estimated soil moisture is maximized and when the improvement is negligible.

Thirdly, during the summer season precipitation in arid or semi-arid regions is often delivered to the ground through high intensity and short duration storms due to strong local convection. ET is also high during daytime. As a result, soil can wet fast and also dry fast. Relying on a daily time step to estimate soil moisture may not provide enough detail to capture the short timescale soil moisture processes and thus induce a large error. To consider the short

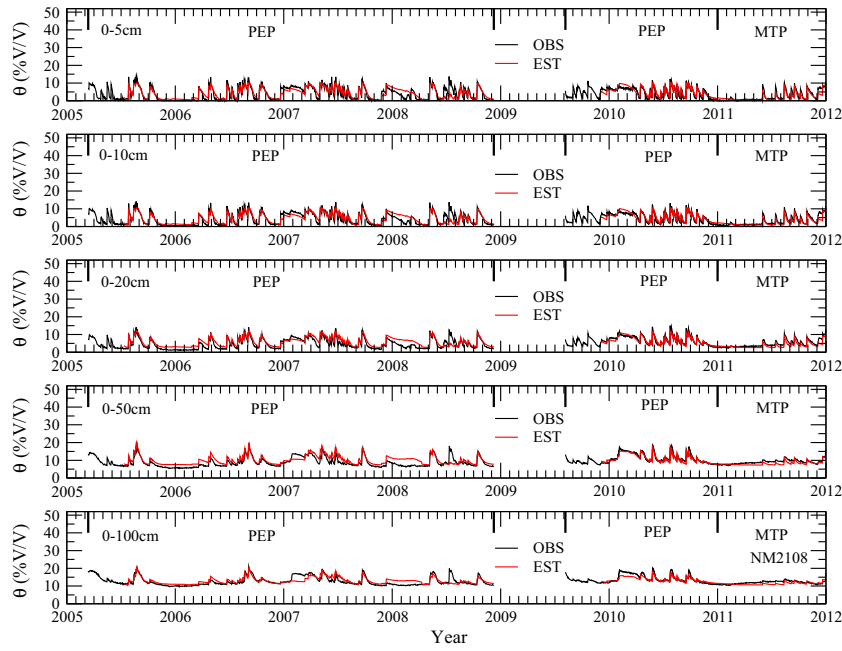


Fig. 10. Observed and estimated soil moisture at NM2108 in 0–5 cm, 0–10 cm, 0–20 cm, 0–50 cm, and 0–100 cm soil columns.

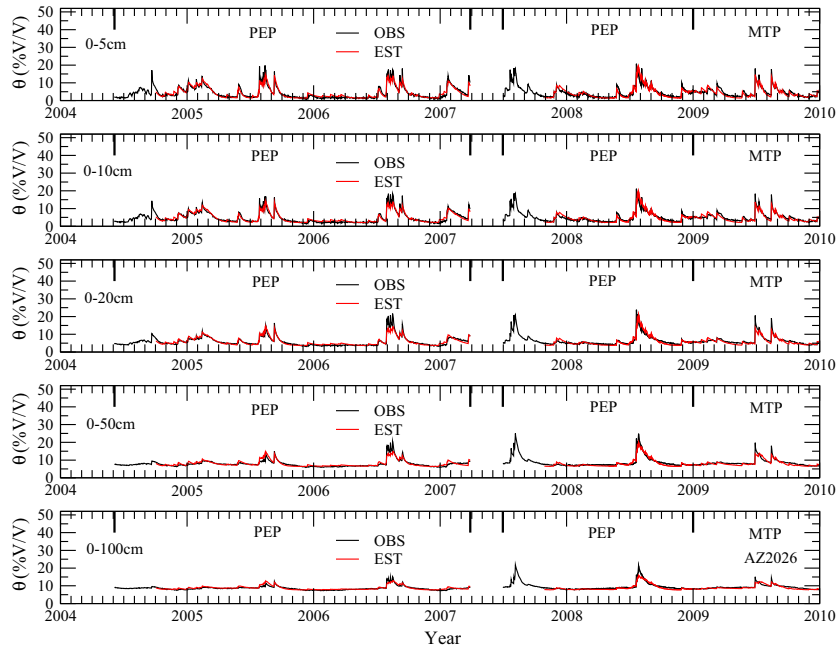


Fig. 11. Observed and estimated soil moisture at AZ2026 in 0–5 cm, 0–10 cm, 0–20 cm, 0–50 cm, and 0–100 cm soil columns.

time scale (e.g., less than one day) variation in soil moisture loss, we can downscale the daily soil moisture loss function to an hourly soil moisture loss function as follows:

where c_8 and c_9 are two parameters, and h is the local time in hours, and $\eta_{i,h}$ is the soil moisture loss coefficient of hour h in day i . Here we assume that the ET is close to zero before sunrise

$$\eta_{i,h} = \begin{cases} c_8 \left\{ c_1 + c_2 \sin \left[\frac{2\pi(\text{DOY}_i + c_3)}{365} \right] + c_5 \left(\frac{T_i}{T_{at}} \right) + c_6 \left(\frac{R_i}{R_{at}} \right) + c_7 LAI \right\} \sin \left[\left(\frac{h - h_{\text{sunrise}}}{h_{\text{sunset}} - h_{\text{sunrise}}} \right) \pi \right] + c_9, & \text{if } h_{\text{sunrise}} \leq h \leq h_{\text{sunset}} \\ c_9, & \text{else} \end{cases} \quad (23)$$

Table 6

Statistics of the estimated soil moisture during the model-testing period.

Site	Depth	Sinusoidal loss function		Hamon PET		JH PET		MB PET	
		RMSE	<i>r</i>	RMSE	<i>r</i>	RMSE	<i>r</i>	RMSE	<i>r</i>
TX	0–5 cm	2.18	0.90	2.13	0.90	2.05	0.90	2.17	0.88
2	0–10 cm	2.02	0.91	1.87	0.90	1.81	0.91	1.92	0.90
1	0–30 cm	1.44	0.89	1.25	0.91	1.24	0.91	1.28	0.91
0	0–50 cm	1.08	0.84	1.01	0.89	1.11	0.87	1.01	0.89
5	0–100 cm	0.972	0.69	1.22	0.73	1.36	0.70	1.17	0.78
NM	0–5 cm	2.72	0.85	3.00	0.81	2.81	0.84	3.15	0.81
2	0–10 cm	2.59	0.85	2.82	0.82	2.60	0.85	3.04	0.80
0	0–20 cm	2.17	0.86	2.34	0.83	2.18	0.85	2.42	0.83
1	0–50 cm	1.61	0.84	1.60	0.81	1.59	0.83	1.64	0.82
5	0–100 cm	1.55	0.74	1.48	0.70	1.55	0.71	1.57	0.71
NM	0–5 cm	1.22	0.92	1.10	0.93	1.06	0.93	0.95	0.94
2	0–10 cm	1.19	0.93	1.10	0.93	1.08	0.93	0.91	0.95
1	0–20 cm	1.62	0.91	2.11	0.89	2.10	0.89	1.91	0.93
0	0–50 cm	0.83	0.82	0.92	0.77	1.01	0.75	0.82	0.82
8	0–100 cm	0.82	0.48	0.94	0.45	1.00	0.40	0.89	0.52
AZ	0–5 cm	1.01	0.93	1.11	0.92	1.19	0.90	1.15	0.91
2	0–10 cm	0.92	0.93	1.07	0.92	1.06	0.91	1.11	0.91
0	0–20 cm	1.00	0.92	1.19	0.90	1.20	0.89	1.21	0.89
2	0–50 cm	0.87	0.89	0.97	0.89	0.97	0.89	1.54	0.80
6	0–100 cm	0.60	0.89	0.71	0.87	0.71	0.87	0.71	0.87

h_{sunrise} and after sunset h_{sunset} . At the hourly time step, intense precipitation can cause significant surface runoff and thus reduce infiltration. To consider surface runoff loss, we can multiply precipitation by one minus a surface runoff coefficient in the equation of B value (see Eq. (9)). The runoff coefficient will be determined based on land surface slope and calibrated through a comparison of observed and estimated soil moistures.

Fourth, in the current soil moisture diagnostic equation, since snowfall and snowmelt processes are not considered, errors in the estimated soil moisture during the winter season could be large especially for the topsoil layer (0–30 cm) if snow accumulation is significant (see Figs. 8 and 9). Including snowmelt in the soil moisture diagnostic equation is not difficult if we have snow water equivalent (SWE) measurements. We can use observed SWE to estimate snowmelt and replace the precipitation term in the equation of B value (Eq. (9)) by the sum of liquid precipitation and snowmelt. 0 °C air temperature can be used to separate rain and snow (e.g., Motoyama, 1990; Lynch-Stieglitz, 1994; Kienzie, 2008).

Fifth, to apply the soil moisture diagnostic equation to estimate soil moisture, estimates of the soil moisture loss function are essential. So far studies related to the soil moisture diagnostic equation (Pan et al., 2003; and Pan, 2012) first used observed soil moisture during a period of time to derive the loss coefficient, and then applied the derived loss coefficient to estimate soil moisture during another period of time. This approach worked well in both previous studies and the current study. However, how to determine the soil moisture loss coefficient without any direct soil moisture measurement is a critical problem for making the soil moisture diagnostic equation more powerful and useful, which requires a systemic study of relationships between the soil moisture loss coefficient and soil, topography, vegetation, climate, and geographical locations. To establish a series of empirical relationships, a large sampling size is necessary. In this study, since we only have 20 samples (e.g., four sites with 5 different soil layers each), no attempt was made to establish such empirical relationships. We argue that after we include snowfall and snowmelt processes in the soil moisture diagnostic equation in a future study, we can use almost all soil moisture data collected at all SCAN sites (about 200 sites in the United States, Puerto Rico and Virgin Islands) to establish relationships between the soil moisture loss coefficient and soil, topography, vegetation, climate, and geographical locations.

Sixth, since the most important use of this soil moisture diagnostic equation is to provide information on soil moisture for farmers to use to determine the quantity and timing of irrigation, it is necessary to expand this research to crop fields. Information on the amount of irrigation water applied and the stage of crop growth will need to be incorporated into the model and tested for reliability. This suggests that a future study on testing the soil moisture diagnostic equation on irrigated croplands is necessary, important, and useful.

5. Conclusions

The method and results presented in this study demonstrate that the soil moisture diagnostic equation has the ability to estimate not only surface soil moisture, but also the entire root-zone soil moisture using only precipitation data. The three advantages associated with the soil moisture diagnostic equation approach identified in previous studies of estimating surface soil moisture using the soil moisture diagnostic equation (Pan et al., 2003; Pan, 2012) still are valid for root-zone soil moisture estimation, i.e., (1) non-cumulative errors in the estimated soil moisture, (2) no regular recalibration is required to correct the cumulative errors, and (3) no numerical iteration and initial moisture inputs are needed. Additionally, this study added another advantage to the soil moisture diagnostic equation approach, namely the flexibility, i.e., soil moisture in any thickness of soil (between 5 and 100 cm in this study) can be estimated using the soil moisture diagnostic equation if the water table is not close to the lower boundary of the soil layer. Both the sinusoidal wave function of the DOY and a linear function of the potential evapotranspiration (PET) were used to approximate the soil moisture loss coefficient. All RMSEs of the estimated soil moistures are less than 3.2 (%V/V), and the accuracy of the estimated soil moistures using the sinusoidal loss function is slightly better than the PET-based loss functions. Although the sinusoidal loss function worked well for the root zone soil moisture, it can be considered as a first order approximation of the soil moisture loss coefficient. To improve the accuracy in representing soil moisture dry-down process, a series of further studies is needed to incorporate temporal variation of air temperature, solar radiation, wind speed, and relative humidity in the soil

moisture loss function. To capture surface runoff and ET at short time scales (e.g., less than one day) in arid and semi-arid regions, an hourly time step may be needed. Considering snowfall and snowmelt processes in the soil moisture diagnostic equation is also important for snow-dominated regions. Finally, estimation of soil moisture in irrigated fields will need to be modeled.

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Appendix A. Calculation of day length

At a location of latitude A° on the earth, the day angle DA (in radians) of the day of year DOY is given by

$$DA = \frac{2\pi(\text{DOY} - 1)}{365} \quad (\text{A-1})$$

The declination of the sun DE (in radians) (Dingman, 2002) of DOY is given by

$$\begin{aligned} DE = & 0.006918 - 0.399912 \cos(DA) + 0.070257 \sin(DA) \\ & - 0.006758 \cos(2DA) + 0.000907 \sin(2DA) \\ & - 0.002697 \cos(3DA) + 0.00148 \sin(3DA) \end{aligned} \quad (\text{A-2})$$

The day length D in hours at the location of latitude A° on DOY is

$$D = \frac{2\text{acos}[-\tan(DE) \times \tan(A\pi/180)]}{\omega} \quad (\text{A-3})$$

where ω is the angular velocity of the earth's rotation and equals to $2\pi/24$ (radian/hr).

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